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MAJOR CONTRIBUTORS TO THE STORY

- **Bhupender Singh**, consultant-Embedded System, Multisoft Systems
- **Sani Theo**, team leader, Efy Labs
- **Seetharaman Devendran**, general manager, MCU, LPRF, sales & marketing, Texas Instruments

Workbench development tools and Sourcery CodeBench development tools (for a limited time). It is compatible with Texas Instruments' Code Composer Studio IDE. Although it has an in-built debugger, it allows one to connect an external debugger through the ARM 20-pin connector, J1.

Kinetis K22 Tower System Module. It features the latest Kinetis K22 32-bit ARM Cortex-M4 MCUs from Freescale and allows USB connectivity with a crystal-less operation. It operates on a stand-alone debugging tool and has an independent battery-operated power supply for real-time clock (RTC). It uses the Kinetis Design Studio IDE along with Freescale MQX real-time operating system for MCU software development along with a Kinetis software development kit.

LPCXpresso. This development board features the LPC1114 Cortex-M0+ MCU from NXP. When used with Eclipse-based IDE and the low-cost target board with an attached JTAG debugger, it can be used to evaluate all the LPC1114 devices. This board supports the GNU C/C++ tool chains available both in free and pro versions. Its compatibility with Arduino UNO and peripheral module (PMOD) standards opens up options for its existing peripheral board.

How to select

It can be a daunting task for anyone selecting the right development board from a plethora of options, having so many features and specifications, available in the market. Luckily, there are some pointers that can help narrow down the choices. Here is how you should proceed with your selection:

1. First select the project to be built and the components required for it.
2. Identify the right MCU with the required number of pins, I/O ports, interface and other features, such as live input power, speed, memory, etc.
3. Create a list of the development

boards that match the project requirements.

4. Identify the availability of the integrated development environment (IDE) and the development tools required for the listed boards.

5. Check the associated simulators and compilers—some boards do not allow GCC compilation.

6. Identify the right board as per the budget of the project.

Although the steps listed above are not the only parameters needed to make a decision, engineers often find these to be useful while selecting a development board for their projects. The idea behind listing these steps is to narrow down the research and select the right development board at an affordable price.

However, Sani Theo, a team leader at Efy Labs says, "Working every now and then with various types of development boards, I feel that the main criteria to select them is to first understand the requirement of the project and to understand what type of MCU will be essential. Then, the other important factor that comes into picture is our familiarity with the MCU. I consider two MCUs—one from Atmel and the other one from some other manufacturer. If I am familiar with Atmel, my approach will be to select the development board based on Atmel MCU."

Bhupender Singh, consultant-Embedded System, Multisoft Systems says, "Identifying the main features on the drivers and the availability of free drivers play an important role when it comes to the selection of a development board." He adds that, currently every manufacturer is tying up with an open source community to come up with an open source development board so that it becomes easy for an engineer to get the associated free software along with free support. ●

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